

ANNOUNCING THE FIRST SIGNAL: POLICY ASPECTS

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Abstract—The detection of an extraterrestrial signal will be a profound event in human history. Procedures for announcing the first signal should be developed with probable global impacts of the announcement in mind. In particular, those procedures should maximize the credibility of the announcement and minimize unwarranted apprehensions. Discussions should begin soon to develop a policy towards announcing the first signal that is acceptable to all groups involved in the search for extraterrestrial intelligence. Experiences in preparing for the announcement of other major events with widespread implications - a nuclear accident and the possible occurrence of a major earthquake - can be usefully examined in order to identify some of the factors that must be considered in developing a policy to announce the first signal.

1. INTRODUCTION

The detection of an extraterrestrial signal will be a profound event in human history. Procedures for announcing to the world that such a signal has been detected and verified should be developed with probable global impacts of that announcement in mind. There almost certainly will be an intense reaction in many sectors of society, in many countries. Questions of perceived security threats are likely to arise. There will be political, social, and philosophical impacts on key institutions: the church, the polity, the media, and cultural, scientific and intellectual organizations. The general public will certainly seek explanations of the meaning and significance of the signal, and various elites are likely to compete in providing such explanations.

The mode chosen to announce the first signal and the context in which that signal is placed by the announcing authority are likely to be critical variables in channeling individual and group reactions in the most positive manner possible. It seems desirable to engage a wide range of people in discussions now, in advance of that detection, about how best to proceed. It also seems desirable to aim toward some form of agreement - an accord, a convention, a protocol, or even a treaty - that embodies the results of such discussions.

This paper is intended as a modest initial contribution to such a consideration. It starts with an assumption that a primary goal of any a priori scheme for announcing the first signal is to minimize confusion, anxiety, fear, and perceptions of threat among the general population regarding the meaning of that announcement. It thus looks to precedents in two other situations that are threatening and could have widespread consequences - earthquakes and nuclear accidents - with respect

to advance agreements on how to announce their occurrence (with respect to nuclear accidents) or their likelihood (with respect to earthquakes). There has been extensive study and debate in both of these areas with respect to how to structure an announcement process, and those interested in developing an approach for announcing the initial extraterrestrial signal may be able to draw some lessons from that experience.

2. ANNOUNCING A NUCLEAR INCIDENT

This past summer, a group of lawyers, nuclear technologists, diplomats, and bureaucrats representing 78 countries and seven international organizations debated in Vienna, under the auspices of the International Atomic Energy Agency (IAEA), two draft global agreements. One of these is to ensure early warning and full information in the event of a nuclear accident with transnational effects; the other provides for international emergency assistance to mitigate the impacts of such an accident. The group reached agreement on the text of conventions with respect to each of these issues, and representatives of most of the 112 members signed the conventions in late September.

The Vienna debate was based on a proposed set of "Guidelines on Reportable Events, Integrated Planning and Information Exchange in a Transboundary Release of Radioactive Materials" that had been developed by a group of experts convened by IAEA[1]. Elements of the expert group recommendations include:

1. defining what constitutes a "reportable event";
2. that states in which such a reportable event occurs have a responsibility to notify a specific national authority (agreed upon in

- advance) in those other states that might be affected by the event;
3. that the notification should be in a language understandable to the receiving states;
 4. that the notifications should include all information required to assess the character and implications of the event. In particular, information on measurement techniques, equipment, and interpretation approaches used to characterize an event ought to be exchanged as part of the advanced planning process.

While specific provisions under each of these categories in the current IAEA debate are probably not germane to the search for extraterrestrial intelligence (SETI) situation, the fact that the need for advanced planning has been recognized, that sustained thought has been given to the requirements of the situation by an appropriate mixture of technical, legal, bureaucratic, and diplomatic experts and that their conclusions have been embodied into a formal convention is certainly of relevance. That such a group was assembled under the auspices of a competent pre-existing international organization is also of interest.

If, as was suggested earlier, it is timely to begin discussions on an *a priori* agreement regarding the announcement of a first signal, then it is important to identify the organizational framework for such discussions, to specify who will have the convening authority, and to begin to identify an appropriate mix of expertise (and experts) to participate in such discussions. Ad hoc occasions such as the current symposium can serve as a catalyst, but are unlikely to be an appropriate framework within which to conduct more sustained discussions and negotiations. It is unlikely that the discussions in the SETI case will include many beyond those active or interested in the search; there is not yet much political salience to the issue, in contrast to the nuclear accident question. A responsible approach to their activities on the part of those engaged in SETI, however, must include attempts, such as this symposium, to broaden the base of individuals and organizations involved.

It should be noted, incidentally, that the fact that there have been several years of discussion at the expert and working levels of a framework for announcing a nuclear accident, in which the Soviet Union has been an active participant, apparently had no effect on Soviet actions following the Chernobyl incident last April. None of the procedures that had been suggested (but not yet agreed upon) were followed. Also,

the debates in Vienna over embodying a set of guidelines into a convention to which governments would subscribe have not been without rancor. Differing views on issues such as the applicability of the convention to defense-related nuclear activity and facilities and the depth of technical information to be provided in advance of an accident made reaching agreement difficult. Similar issues on content and adherence are unlikely to be involved in the process of developing a SETI agreement.

3. ANNOUNCING AN EARTHQUAKE PREDICTION

One aspect in which the situation with respect to nuclear accidents and with respect to an initial extraterrestrial signal is likely to be different is certainly associated with the event. An accident is a relatively unambiguous occurrence, while the detection of a first signal is likely to be accompanied by a statement of the level of confidence in the detection. In this respect, the situation is closely parallel to that associated with earthquake predictions, in which increasing scientific knowledge and technological capabilities create the possibilities of lowering uncertainties associated with predicting an earthquake to a level on which public and private actions could be based. Like those engaged in SETI, scientists and authorities considering when to make public an earthquake prediction have to make a difficult judgement of how sure they are of the validity of the prediction before announcing it.

As earthquake prediction capabilities have improved over the past fifteen years there has been a substantial amount of attention paid to the policy dimensions of this capacity[2]. Some of the information and insights from these discussions appear relevant to the SETI situation.

It seems appropriate to characterize current SETI activities around the world as research, not only into the existence of other intelligences, but also into how to detect such intelligences. Given the unavoidable uncertainty about the potential nature of any one communication or unintended signal, and given the possibility of a large variety of such signals, there clearly is no way to specify one right way to conduct the search. Thus there is likely to be a fair degree of uncertainty associated with any detection, and that uncertainty has to be accounted for in deciding how to announce a signal, and what explanations to include with that announcement.

There is a problem that has troubled those developing capabilities for earthquake

predictions, since the prediction itself can have widespread economic, social, psychological, and political consequences. Lambright and Heckley[3] have noted that "prediction has proved difficult to 'sell' to many potentially affected communities," both because of uncertainty about the validity of the predictions and because "some interests - such as banks and real estate firms - see prediction more as a threat than opportunity for them." Those developing an approach to announcing the first signal will need to identify those interests unlikely to welcome such an announcement (most orthodox religions, e.g.) and investigate whether there are ways to lower resistance to such an announcement.

Two countries in which earthquake prediction technology has been under active development and which have put in place policies for earthquake predictions are the United States and Japan (China might be included in this list; a successful 1973 prediction there is said to have saved tens of thousands of lives. Insufficient information on Chinese policies regarding earthquake predictions was available to the authors of this paper). The U.S. and Japanese approaches to announcing a prediction have both similarities and differences that deserve attention.

The Japanese approach is hierarchical and tightly controlled. The Japanese have an active earthquake research program and have deployed instruments to detect the precursors of an anticipated very large earthquake. A standing committee of scientists meets at least monthly to review data from these instruments and elsewhere; there is an active attempt to develop a prediction. If this group concludes that a prediction is warranted, it transmits this information in confidence to the head of a technical agency of the Japanese government, who examines the evidence on which the prediction is based. He then transmits the prediction and his advice regarding it to the Prime Minister. The Japanese media have agreed to a brief moratorium on news at this point, so that the Prime Minister can issue an official warning, if he chooses to do so, prior to any unofficial news of the prediction. In order to discourage excess scientific conservatism, the Japanese government has indicated its willingness to deal with the economic and social consequences of a few incorrect predictions[4].

The U.S. approach to dealing with an earthquake prediction is much more disjointed and incremental. A mixture of federal, state, and local interests and authorities are involved,

with no comprehensive planning on how they will interact. Most activity in the United States has focused on Southern California, where the existence of the San Andreas Fault makes a severe earthquake at some point rather likely. The interest in earthquakes in this region and the open and pluralistic nature of American society make it almost impossible to control the announcement of a prediction as is done in Japan. As one report noted:

An earthquake prediction can come from scientists at public or private universities, from State or Federal scientists, non-affiliated scientists, or individuals using non-scientific or semi-scientific prediction techniques....A prediction will become public knowledge and be broadcast by the news media very quickly. Experience has demonstrated that a possible prediction is likely to be uncovered by the press unless it is announced to the public promptly. Initially, public response to a prediction will depend on the credence and publicity given to that prediction by the news media[5].

Both California and the national government in the United States have put in place elements of a system for handling earthquake predictions. There exist two groups that evaluate earthquake predictions, the California Earthquake Prediction Evaluation council (CEPEC) and the National Earthquake Prediction Evaluation Council (NEPEC). These operate independently of one another, and are passive recipients and evaluators of predictions emanating from other sources that are convened on an ad hoc basis, rather than standing organizations actively attempting to predict earthquakes. This form of independent verification (or denial) of the validity of a prediction seems desirable in principle, but in practice the verification process has operated slowly and thus has followed, rather than preceded, public knowledge of a prediction.

For example, in 1976 Cal Tech scientist James Whitcomb predicted an earthquake in the Los Angeles area. It took CEPEC several months to declare this prediction invalid; in the interim, there was substantial interest in the prediction and controversy about how to respond to it. When CEPEC declared the prediction invalid, Whitcomb was "the focus of much undesired controversy, ridicule, and even threats of law suits[6]." This experience may be educational for those scientists eager to go public with what they think is a detected extraterrestrial signal. It seems desirable to work towards a rapid system

of independent verification of a detection, so that there is some chance of avoiding widespread public knowledge of possible signals that the SETI community ultimately judges to be something other than of extraterrestrial origin.

Having a prediction publicized before it is validated can make it difficult to reverse initial reactions. For example, in 1980 a U.S. researcher predicted, and announced, that a major earthquake would occur near Lima, Peru in August 1981. The Peruvian government sought the assistance of the U.S. NEPEC, which indicated that the prediction was based on speculative and vague evidence and should be disregarded. Despite widespread publicity in Peru given to this disclaimer and an August 1981 visit to Peru by the vice-chairman of NEPEC, many in the country continued to give credence to the prediction; many left the Lima vicinity on the day one of the foreshocks was supposed to occur[7].

There have been some attempts to begin a program of international cooperation in drawing up guidelines for the formulation, evaluation, and communication of earthquake predictions; these discussions were begun early in this decade under the auspices of the International Association of Seismology and Physics of the Earth's interior[8]. The results of these discussions could be of interest to those interested in developing an agreement for announcing the first signal. In particular, the fact that the discussions are being carried out within the framework of a scientific union may suggest candidates for organizing the SETI dialogue.

4. CONCLUSION

Given the fact that SETI is an activity in which many scientists in several countries are involved, it is likely that the pluralistic U.S. experience in dealing with earthquake predictions is more relevant than the hierarchical Japanese approach. The problems of dealing with a conclusion that is accompanied by some degree of uncertainty, which is relevant to broad societal interests and values, and about which there is widespread interest are evident in the earthquake prediction situation. These are the factors that will have to be taken into account as the SETI community begins the necessary process of developing plans for announcing an initial detection of a signal of extraterrestrial origin. This paper has attempted to suggest some precedents and experience upon which that discussion can draw.

REFERENCES

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